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Statistics project: The influence of average income per employee over expenses per trip for the year 2023

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1. INTRODUCTION

This study looks at how people's income affects tourism spending in Europe. Specifically, it examines whether the average yearly salary in a country influences how much people spend on short tourism trips lasting 1–3 days, both domestically and internationally. The research is based on John Maynard Keynes's Absolute Income Hypothesis, which suggests that people spend more money when their income increases. In tourism, short trips are considered discretionary spending, meaning they are more likely to increase when people earn higher wages. Understanding this relationship is important for tourism managers, hotels, and destination marketing organizations because it can help predict travel spending patterns and support business planning.

Using data from 26 European countries in 2023, the project applies statistical and econometric methods to test whether average national income is a significant predictor of short-term tourism expenditure.

2. DATABASE AND VARIABLES

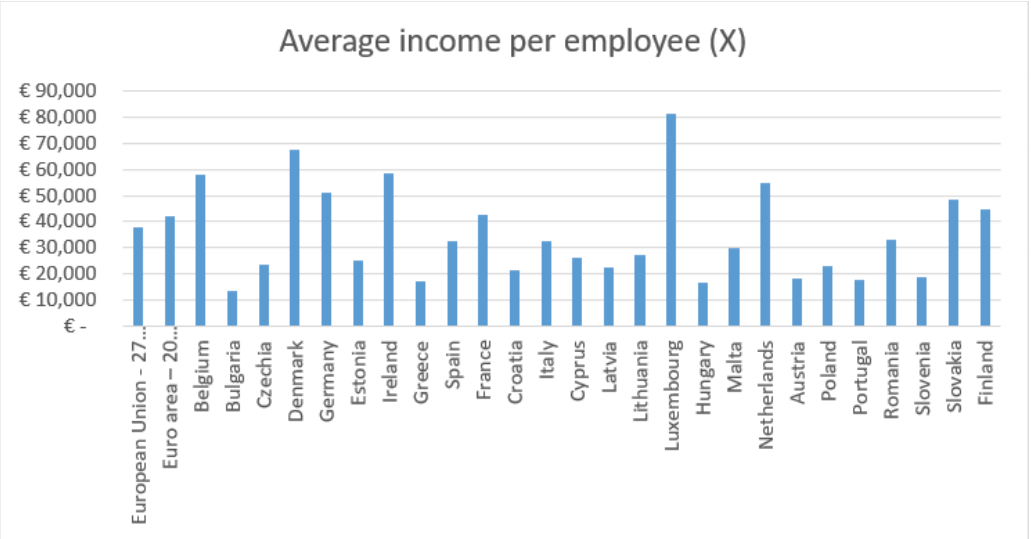
A cross-sectional dataset was extracted from the Eurostat official database for the year 2023. The framework isolates two specific indicators:

- ***Independent Variable (X)*:** *Average income per employee*, measured in Euros (€) per annum. This measures the purchasing power of the domestic workforce.

The bar chart shows large differences in average annual income across the European countries included in the study.

Countries such as Luxembourg, Denmark, Ireland, and Belgium report the highest income levels. Lower average incomes are observed in countries such as Bulgaria, Greece, Hungary, and Portugal. The chart suggests a clear economic gap between higher-income Western/Northern European countries and lower-income Eastern/Southern European

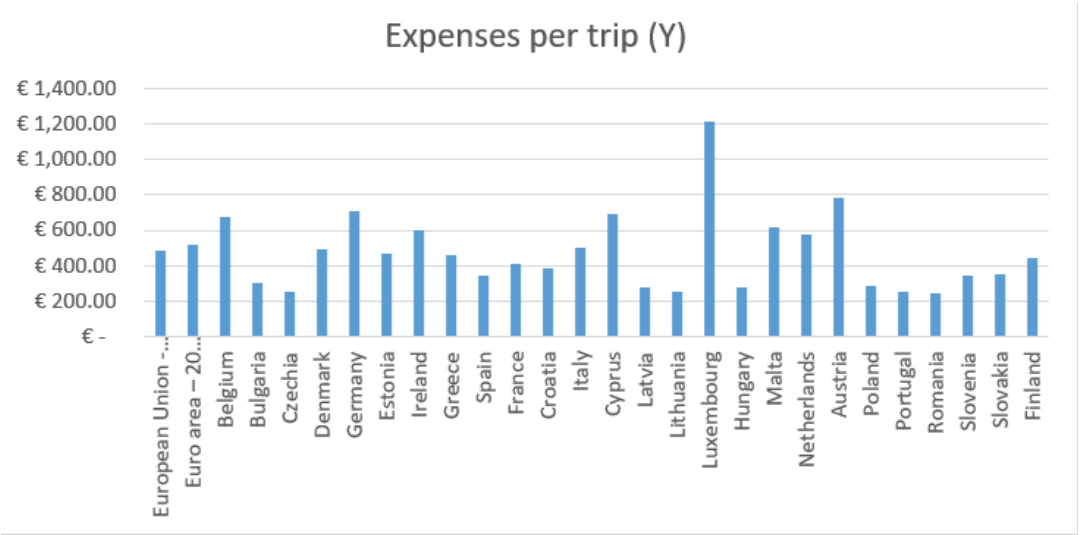
countries. Income distribution appears uneven, with a few countries having exceptionally high wage levels compared to the rest of the sample.



- ***Dependent Variable (Y):*** *Expenses per trip (1–3 days)*, measured in Euros (€) per individual trip. This variable measures the average amount spent per individual short-term tourism trip.

The tourism expenditure chart also shows variation in spending behavior across countries.

Countries with relatively high travel spending include Austria, Germany, Italy, and Netherlands. Luxembourg records the highest average tourism spending per trip by a large margin. Lower travel expenses are observed in Bulgaria, Romania, Portugal, and Hungary.The variation in trip expenses suggests that tourism spending differs significantly between countries.



The database used in this study:

GEO (Labels)	Expenses per trip (Y)	Average income per employee (X)
European Union - 27 countries	€ 484.19	€ 37,863
Euro area – 20 countries	€ 515.30	€ 41,848
Belgium	€ 673.09	€ 57,989
Bulgaria	€ 303.84	€ 13,503
Czechia	€ 252.33	€ 23,454
Denmark	€ 488.73	€ 67,604
Germany	€ 705.28	€ 50,998
Estonia	€ 467.16	€ 24,899
Ireland	€ 598.47	€ 58,679
Greece	€ 458.27	€ 17,013
Spain	€ 346.72	€ 32,587
France	€ 406.75	€ 42,662
Croatia	€ 384.41	€ 21,523
Italy	€ 501.82	€ 32,749
Cyprus	€ 689.41	€ 26,430
Latvia	€ 278.88	€ 22,293
Lithuania	€ 250.22	€ 27,178
Luxembourg	€ 1,213.83	€ 81,064
Hungary	€ 273.96	€ 16,895
Malta	€ 619.30	€ 29,989
Netherlands	€ 573.35	€ 54,508
Austria	€ 784.89	€ 18,054
Poland	€ 284.39	€ 22,933
Portugal	€ 251.68	€ 17,739
Romania	€ 246.80	€ 33,081
Slovenia	€ 345.91	€ 19,001
Slovakia	€ 350.32	€ 48,391
Finland	€ 442.08	€ 44,619

3. DESCRIPTIVE STATISTICS AND THE EMPIRICAL RULE

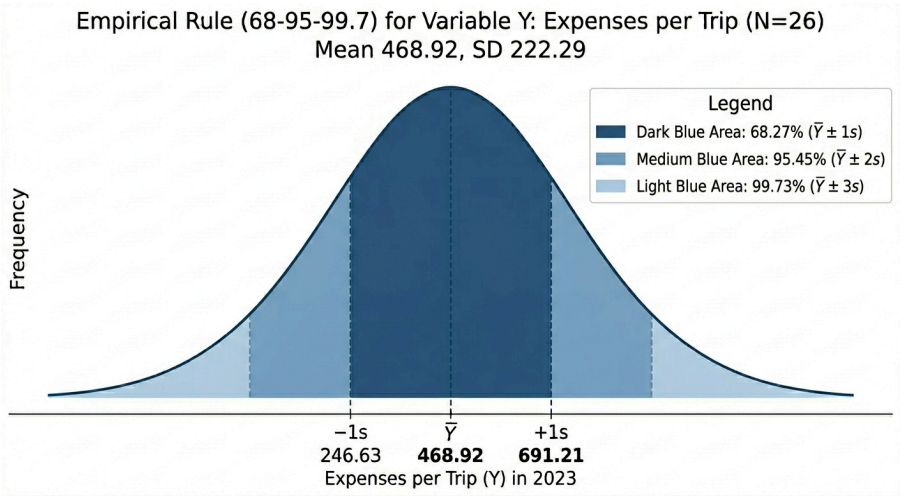
Before creating prediction models, we first examine the basic patterns in the data. Using data from 26 observations and by using Excel functions, the main statistical measures for the dependent variable *Expenses per trip (Y)* are:

Sample mean ($\bar{Y}$): 468.92, Observations (N): 26, Standard deviation (σ): 222.29

To model the distribution pattern, we apply the Chebyshev-Empirical criteria, assuming an approximately symmetrical distribution:

- ***68% Rule ($\bar{Y} \pm 1\sigma$):*** The interval is **[246.63, 691.21]**. In the database, 23 out of 26 countries (88.46%) fall within this range.
- ***95% Rule ($\bar{Y} \pm 2\sigma$):*** The interval is **[24.34, 913.50]**. In the database, 25 out of 26 countries (96.15%) fall within this range.
- ***99.7% Rule ($\bar{Y} \pm 3\sigma$):*** The interval is **[0, 1135.79]**. In the database, 25 out of 26 countries (96.15%) fall within this range.

The distribution appears relatively concentrated around the mean, with most observations falling within one standard deviation. However, because the sample size is limited, no definitive conclusion regarding kurtosis or normality can be made without additional tests.



4. HYPOTHESIS TESTING FOR THE POPULATION MEAN

To determine whether the sample represents an accurate reflection of European short-term trip expenditures, we run a parametric two-tailed hypothesis test. We evaluate whether the true population mean expenditure (μ) differs significantly from a cited reference value of $\mu_0 = 400€$

Step 1: Formulation of Hypotheses

- $H_0 : \mu = 400€$ (The true mean expenditure does not differ from the 400€).
- $H_1 : \mu \neq 400€$ (The true mean expenditure is significantly different from 400€).

Step 2: Test Statistic Selection

Because the population standard deviation (σ) is unknown and our sample size is smaller than 30, we utilize the t-statistic distribution:

- Standard error (SE) : $\frac{S}{\sqrt{n}} = \frac{222.92}{5.09902} = 43.7182$
- $T_{calculated} = \frac{(X-\mu)}{s/\sqrt{n}} = \frac{468.92-400}{43.7182} = 1.576$

Because $T_{calculated} = 1.576$ is less than the critical value $T_{crit} = 2.0595$, we fail to reject the null hypothesis (H_0). There is insufficient empirical evidence at the 95% confidence level to state that the true mean expenditure per short-term trip across Europe varies significantly from the 400€ baseline.

5. LINEAR REGRESSION AND CORRELATION ANALYSIS

REGRESSION

To analyze the relationship between the independent and dependent variables , we formulate a classic bivariate ordinary least squares linear regression model:

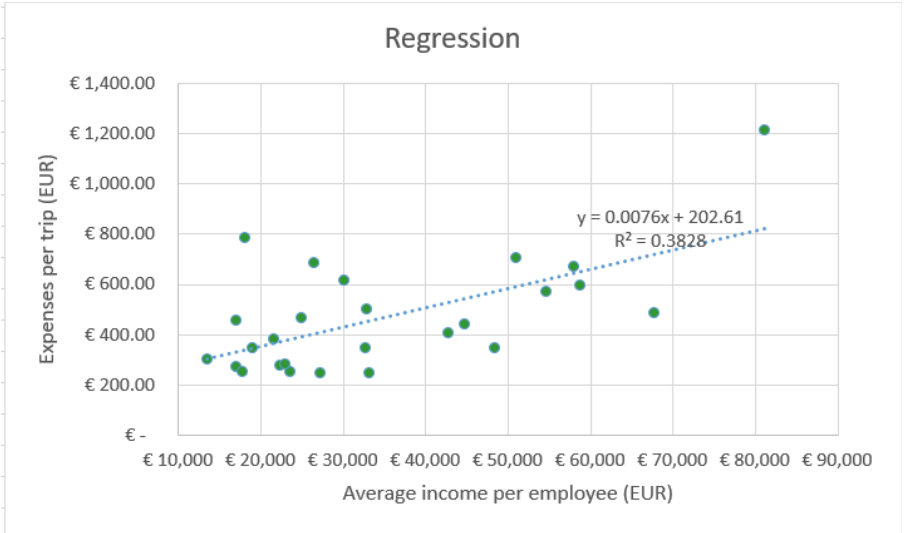
$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

Regression Statistics									
Multiple R	0.618693965								
R Square	0.382782223	38.27							
Adjusted R Square	0.357064815								
Standard Error	178.2411201								
Observations	26								
ANOVA									
	df	*SS*	*MS*	*F*	*Significance F*				
Regression	1	472868.4957	472868.5	14.88417	0.000753764	< 0,05			
Residual	24	762477.5252	31769.9						
Total	25	1235346.021							
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95,0%*	*Upper 95,0%*	
Intercept	202.6125938	77.373429	2.618633	0.015056	42.92168502	362.3035027	42.92168502	362.3035027	
X Variable 1	0.007643735	0.001981268	3.858001	0.000754	0.003554598	0.011732872	0.003554598	0.011732872	

The Pearson correlation value (Multiple R) = 0.61869 confirms a moderate to strong positive linear relationship. As national income increases, travel expenditure also tends to increase.

Based on the regression results, the estimated empirical model is:

$$\hat{Y} = 202.6121 + 0.00764 \cdot X$$



The scatter plot provides a visual representation of the relationship between average income and travel expenses across the countries. Each data point represents a specific country, and their distribution shows a clear upward trend.

The fact that the points are relatively close to the trendline supports our R square value of 0.3838, indicating that while other factors influence travel costs, income appears to be an important factor influencing tourism expenditure. The positive slope of the line visually

confirms that as national average income rises, individual spending on short-term trips increases proportionally. The regression equation displayed on the chart, y = 0.0076x + 202.61, provides the specific predictive model for this relationship, indicating that for every additional euro of average income, trip expenses are expected to increase by approximately 0.0076 euro.

EVALUATION OF THE MODEL:

The R Square value of 0.38278 indicates that exactly 38.28% of the variation in tourism expenditure is associated with variations in average employee income. The remaining 61.72% of variance cannot be explained by income alone and is absorbed by the error term. This unexplained variance is driven by external variables omitted from this simple model, such as:

- Geographic proximity and availability of budget airline
- Domestic inflation differentials and local cost-of-living within the destination.
- Distinct cultural variances regarding consumer saving rates versus leisure allocation behaviors.

ANOVA

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To determine if the model is valid, we test the following hypotheses:

- H_0**:** $\beta 1 = 0$ (The true population slope is zero; national average income holds zero predictive value over short-term vacation spending).
- H_1**:** $\beta 1 \neq 0$ (The true population slope is not zero; national average income is a statistically significant predictor of travel spending).

To make a decision at the standard significance level $\alpha = 0.05$, we evaluate the significance F provided in the sheet, which equals 0.000753764.

Because the Significance F is vastly lower than 0.05, we reject the null hypothesis (H_0). Therefore, we conclude with a high level of confidence that the regression model is highly valid, and an employee's average income is a reliable structural predictor of short-duration holiday expenditures inside the European Union.

6. MANAGERIAL IMPLICATIONS

In conclusion, the analysis indicates that average employee income is positively associated with short-term tourism expenditure across the selected European countries. The regression results show that higher income levels tend to correspond to higher travel spending, although additional factors also influence tourism behavior. Future research could improve the explanatory power of the model by incorporating additional variables such as inflation, transportation accessibility, and regional tourism costs.